

**Contribution of the Late Positive Potential to Emotional Memory: A Joint Neuro-Cognitive
Approach to Memory Prediction**

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Computational modeling of the Early-LPP contribution to emotional recall

The behavioral and EEG analyses identified two forms of variation in recall. At the category level, negative items were recalled more often than neutral items. At the item level, Early LPP was more strongly associated with subsequent recall for negative than neutral items. The empirical analyses distinguish these patterns statistically, but they do not determine how trial-level Early LPP should be mapped onto the encoding mechanism proposed to produce emotional memory enhancement. We therefore tested whether using item-level Early LPP to scale learning strength could replace a category-wide increase in learning strength, or whether the two forms of modulation explained complementary variation in which items were recalled.

We organized this comparison around three focused questions. We tested (1) whether preferential learning of negative items improved prediction of which items were recalled; (2) whether scaling learning with item-level Early LPP could replace that category-wide advantage or improve prediction after the advantage was included; and (3) whether Early-LPP scaling was better applied to both negative and neutral items or only to negative items. We therefore crossed the presence or absence of preferential emotional learning with three LPP specifications: no LPP-based learning modulation, one relationship shared across negative and neutral items, or one relationship limited to negative items. This crossing produced six models. All six models used the same memory architecture. We first describe that shared architecture and show where preferential emotional learning and Early LPP altered learning strength. We then describe model fitting, compare the six models, and use simulations to determine which observed patterns the fitted models reproduced.

eCMR Model Specification

We modeled recall with an adaptation of emotional Context Maintenance and Retrieval (eCMR) (Talmi et al., 2019). Like other retrieved-context models (Howard & Kahana, 2002; Polyn et al., 2009), eCMR represents context as an evolving internal state shaped by recently

processed events. A studied item changes this state and forms associations in both directions: the item can later reinstate its study context, and the current context can later support retrieval of the item. Because a recalled item reinstates context before the next recall, stronger context-to-item learning can change which items enter the recalled set.

The context representation contained temporal and source components,

$$\mathbf{c}_i = \begin{bmatrix} \mathbf{c}_i^T \\ \mathbf{c}_i^S \end{bmatrix}.$$

Temporal context represented the evolving sequence of events, whereas source context represented whether an item was negative or neutral. These were components of one context representation rather than separate memory systems. All six models represented both source categories; models without preferential emotional learning therefore still distinguished negative from neutral items.

Within each list, a feature vector $\mathbf{f}_i$ uniquely identified each studied item i . Two associative mappings connected item features and context. The feature-to-context mapping M^{FC} allowed an item to reinstate context, whereas the context-to-feature mapping M^{CF} allowed context to support candidate items. On each study trial, the item first generated temporal and source contextual inputs. For either component, $\mathbf{c}_i^{\text{IN}}$ denotes the input generated by item i , and context updated as follows:

$$\mathbf{c}_i = \rho_i \mathbf{c}_{i-1} + \beta \mathbf{c}_i^{\text{IN}}.$$

Here, β controlled how strongly the current item changed context, and ρ_i kept context at unit length. Temporal and source context used separate encoding-drift parameters. Only after both components had been updated did the model learn the item's associations, binding the item to a context state it had already helped create.

Feature-to-context learning stored the updated context that the item could later reinstate. Context-to-feature learning determined how strongly that context would later support the item.

The temporal and source blocks of M^{CF} , M_T^{CF} and M_S^{CF} , were updated as follows:

$$\Delta M_{T,i}^{CF} = \phi_i \mathbf{f}_i(\mathbf{c}_i^T)^\top, \quad \Delta M_{S,i}^{CF} = S_{ij}^S \mathbf{f}_i(\mathbf{c}_i^S)^\top.$$

The primacy gradient $\phi_i = 1 + \phi_s \exp[-\phi_d(i - 1)]$ temporarily strengthened learning for early list positions. The focal emotional-learning and Early-LPP multipliers entered only the source learning strength S_{ij}^S , allowing them to be compared through the same pathway while temporal learning remained unchanged.

At recall onset, both context components shifted toward their start-of-list states. Temporal and source context then provided additive support a_i for each candidate item:

$$a_i = [M^{CF} \mathbf{c}]_i = [M_T^{CF} \mathbf{c}^T]_i + [M_S^{CF} \mathbf{c}^S]_i.$$

Among the studied items not yet recalled, choice sensitivity τ_c converted this support into the probability of selecting item i next:

$$P(R_t = i \mid \mathcal{A}_t) = \frac{a_i^{\tau_c}}{\sum_{k \in \mathcal{A}_t} a_k^{\tau_c}}, \quad i \in \mathcal{A}_t,$$

where $\mathcal{A}_t$ contains the items still available at recall step t . After each selection, the item was removed from this set and reinstated its temporal and source context, changing the cue for the next recall. This cycle links source learning strength to an item's probability of entering the recalled set.

The fitted model omitted item-specific semantic similarities and the source-switch disruption used in published eCMR; recall order was unavailable to diagnose the contributions of semantic similarity and source switching to retrieval dynamics. It also fixed source reinstatement at its maximum and replaced the published accumulator and stopping process with normalized selection among unrecalled items. The likelihood consequently conditioned on each list's observed recall count rather than predicting when recall stopped. These fixed departures were shared across all six models, which differed only in their scaling of source learning strength.

Encoding-strength hypotheses

Every model estimated a source-learning baseline $\omega\phi_i$, where ω scaled source relative to temporal learning and ϕ_i supplied the primacy gradient. Preferential emotional learning multiplied this baseline by $\phi_{\text{emot}}^{e_i}$, where $e_i = 1$ for negative items and $e_i = 0$ for neutral items. Models without this modulation fixed $\phi_{\text{emot}} = 1$; the remaining models estimated it over $(0, 10]$, allowing either enhancement or attenuation of negative-item learning. Because this category-wide multiplier cannot distinguish among negative items, Early LPP supplied the item-level variation.

Because our modeling target was which items entered each list’s recalled set, conditional on the observed recall count, we represented Early LPP as a within-list deviation:

$$z_{ij} = \text{EarlyLPP}_{ij} - \overline{\text{EarlyLPP}_j}.$$

Thus, z_{ij} indicates whether item i elicited a larger or smaller Early LPP than its listmates. The Early-LPP coefficients test whether this relative difference predicts entry into the recalled set. This centered predictor deliberately excludes between-list mean Early LPP; testing whether lists with larger mean Early LPP produce more recalls would require retaining the list mean as a separate predictor and modeling recall count. Because the preceding empirical regression did not separate within- and between-list variation, its Early-LPP coefficients are not numerical counterparts of the computational Early-LPP coefficients.

Source learning strength was the product of the ordinary source-learning baseline, the category-wide emotional-learning multiplier, and an item-level Early-LPP multiplier:

$$S_{ij}^S = \omega\phi_i \times \phi_{\text{emot}}^{e_i} \times \exp[\kappa z_{ij} + \kappa_{\text{emot}} e_i z_{ij}].$$

Within the Early-LPP multiplier, κ scaled source learning for both negative and neutral items, whereas κ_{emot} added negative-item-specific scaling. The exponential link kept learning strength positive and made Early-LPP effects proportional; when $z_{ij} = 0$, the Early-LPP multiplier equaled 1 and recovered the category-level baseline. No-LPP models fixed both coefficients at zero. General-LPP models estimated κ alone, whereas equal-complexity Emotion-dependent-LPP

models estimated κ_{emot} alone. The empirical Emotion $\times$ Early-LPP interaction motivated the General-versus-Emotion-dependent comparison, which determined whether the single Early-LPP coefficient better predicted recalled sets when applied to both categories or only to negative items.

Model-comparison design

Crossing a fixed or freely estimated emotional-learning multiplier with the three Early-LPP mappings produced the six models in Table 1.

Within each row, the two columns test preferential emotional learning while holding the Early-LPP mapping fixed. Within each column, the LPP models test whether the neural measure adds information beyond the corresponding No-LPP model. The General and Emotion-dependent rows compare two equal-complexity placements of one Early-LPP coefficient. The 12-parameter General-LPP and Emotion-dependent-LPP models can also be compared directly with the 12-parameter Preferential-learning model, testing whether LPP-derived scaling can replace a free category-wide multiplier.

Table 2 summarizes the parameters and the constraints that created the six cells. Eleven common parameters governed context updating, associative learning, primacy, and recall choice; each optional learning multiplier added one parameter.

The grid estimates one Early-LPP coefficient at a time and therefore does not test for a residual General slope after an Emotion-dependent slope is included. It also holds the source-context pathway fixed and cannot establish that the predictive Early-LPP relationship is specific to source rather than temporal learning. Having defined the alternatives, we next describe the common evidence used to compare them.

Pooled estimation and evaluation

The modeling dataset matched all 6,840 study events to recall outcomes and the Early-LPP values described in the preceding EEG methods. Values missing for 371 events (5.4%) were imputed without using recall status: first from the same item category on the participant's list, then from the complete list, and finally from the dataset-wide category mean. Each model used one parameter vector shared across all 342 lists from 38 participants; the analysis therefore estimates a

pooled pattern rather than participant-specific mechanisms. Each participant contributed only nine lists, too few independent list histories to support stable estimation of 11 to 13 parameters.

The records identified each list's recalled items but not their output order. Let $n_j = |\mathcal{S}_j|$ be the number of items in recalled set $\mathcal{S}_j$, let π_{jk} be its k th sampled order, and let θ be the model parameters. Because reinstatement makes eCMR sequence-dependent, we approximated the evidence for each set by averaging the probabilities of $K = 200$ fixed sampled orders:

$$\tilde{P}(\mathcal{S}_j \mid \theta, n_j) = \frac{1}{K} \sum_{k=1}^K \prod_{t=1}^{n_j} P(R_{jt} = \pi_{jk,t} \mid \pi_{jk,1:(t-1)}, \theta) .$$

Each product is one possible recall sequence and the outer average combines the sampled sequences. The omitted factor $n_j!$ depends only on the observed set size, so it adds the same constant to every negative log-likelihood (NLL) and does not change parameter estimates or model rankings. Every model used the same sampled orders and random seed. Because the likelihood also excluded termination, it predicts recalled-item identity conditional on the observed number of recalls rather than explaining how many items were recalled.

We minimized summed NLL with differential evolution ([Storn & Price, 1997](#)), using 15 candidate vectors per free parameter for at most 1,000 generations. Three independent starts were run for each model, and the lowest NLL was retained. Early-LPP coefficients were constrained to $[0, 0.2145]$ in accordance with the directional hypothesis; the upper bound permitted a tenfold multiplier at the largest observed centered Early LPP. We compared best-found solutions using $AIC = 2NLL + 2p$, where p is the number of free parameters ([Burnham & Anderson, 2002](#)).

To interpret the fit differences, we generated 200 complete datasets from each fitted model using the observed lists, Early-LPP values, and recall counts. We summarized the negative–neutral recall difference and the remembered–forgotten Early-LPP difference within each category. The Early-LPP diagnostic used the original z-transformed values for continuity with the empirical analysis, although within-list-centered values modulated learning. These summaries were consequences of which items the model recalled, not additional fitting targets, and therefore provide explanatory diagnostics rather than out-of-sample validation.

Model-comparison results

Overall comparison

The model combining preferential emotional learning with Emotion-dependent Early-LPP modulation had the lowest AIC (Table 3). Its best-found NLL was 8241.988 with 13 free parameters, yielding an AIC of 16509.977; the planned contrasts below explain this ranking.

The fit statistics compare conditional prediction, whereas the simulations show which empirical patterns accompanied the fit differences. The figures contrast the observed data with preferential emotional learning alone, Emotion-dependent Early-LPP modulation alone, and their combination; the tables report all six models.

Contribution of preferential emotional learning

Freeing the emotional-learning multiplier lowered AIC under every Early-LPP specification: by 108.494 points with no LPP, by 96.287 points with General LPP, and by 80.133 points with Emotion-dependent LPP. The fitted multiplier exceeded 1 in all three models. Across all six models, those with preferential emotional learning reproduced the observed negative-minus-neutral recall difference (predicted 0.119–0.140; observed 0.124), whereas models without it produced differences of only 0.000–0.019 (Figure 1).

Contribution of Early LPP

Without preferential emotional learning, General LPP lowered AIC by 21.971 points and Emotion-dependent LPP lowered it by 56.693 points relative to the source-context baseline. With preferential emotional learning, the corresponding AIC improvements were 9.764 and 28.332 points. Thus, Early LPP improved conditional prediction whether the emotional-learning multiplier was fixed or estimated.

The Preferential-learning model had an AIC 86.523 points lower than the General-LPP model and 51.801 points lower than the Emotion-dependent-LPP model; all three models had 12 free parameters. Neither LPP mapping therefore replaced the category-wide multiplier, even though both added predictive information to models with or without it.

In the observed data, remembered items had a 0.455 larger mean Early-LPP z score than forgotten items within the negative category, whereas the corresponding neutral-item difference was 0.035. Models without Early-LPP modulation produced little remembered-versus-forgotten separation in either category. Because simulations kept each item's Early-LPP value fixed, the separation in Figure 2 arose only when a model preferentially recalled items with larger values. Early-LPP modulation therefore supplied item-level information absent from the category-wide multiplier.

General versus Emotion-dependent Early LPP

Emotion-dependent LPP lowered AIC by 34.723 points relative to General LPP when preferential emotional learning was absent and by 18.568 points when preferential emotional learning was present. Because the alternatives had equal numbers of parameters, these AIC differences reflected fit rather than complexity penalties.

General LPP separated remembered and forgotten items in both categories, whereas Emotion-dependent LPP concentrated the separation among negative items. With preferential emotional learning, the predicted difference between the negative-item and neutral-item remembered–forgotten Early-LPP contrasts was 0.264 under General LPP and 0.424 under Emotion-dependent LPP, compared with 0.420 observed. The General Early-LPP coefficient was below its upper bound ($\hat{\kappa} = 0.151$ of 0.2145), so the preference for Emotion-dependent LPP was not attributable to the General coefficient pressing against its declared bound.

The preferred model combined preferential emotional learning with Emotion-dependent Early-LPP modulation. It estimated a source-learning scale of 2.669, an emotional-learning multiplier of 1.577, and an emotion-dependent log-learning slope of 0.115. Because these parameters were estimated jointly with the remaining model parameters, their isolated magnitudes should not be assigned direct psychological meanings.

Summary and scope

Taken together, the comparisons support complementary roles for preferential emotional learning and Early-LPP modulation. The category-wide multiplier reproduced the negative-item

recall advantage, whereas Emotion-dependent Early LPP distinguished which negative items were recalled. Their combination was preferred, implementing the proposed pathway from Early LPP through source learning strength to the conditional probability of entering the recalled set. LPP-derived scaling therefore complemented rather than replaced preferential emotional learning.

Supplementary material

Complete pooled parameter estimates

Table 5 defines the compact model codes used in Table 6.

- Burnham, K. P., & Anderson, D. R. (2002). *Model selection and multimodel inference: A practical information-theoretic approach* (2nd ed.). Springer. <https://doi.org/10.1007/b97636>
- Howard, M. W., & Kahana, M. J. (2002). A distributed representation of temporal context. *Journal of Mathematical Psychology*, 46(3), 269–299. <https://doi.org/10.1006/jmps.2001.1388>
- Polyn, S. M., Norman, K. A., & Kahana, M. J. (2009). A context maintenance and retrieval model of organizational processes in free recall. *Psychological Review*, 116(1), 129–156. <https://doi.org/10.1037/a0014420>
- Storn, R., & Price, K. (1997). Differential evolution – a simple and efficient heuristic for global optimization over continuous spaces. *Journal of Global Optimization*, 11(4), 341–359. <https://doi.org/10.1023/A:1008202821328>
- Talmi, D., Lohnas, L. J., & Daw, N. D. (2019). A retrieved context model of the emotional modulation of memory. *Psychological Review*, 126(4), 455–485. <https://doi.org/10.1037/rev0000132>

Table 1

Six models formed by crossing Early-LPP modulation with a fixed or estimated emotional-learning multiplier. All models contained temporal and negative/neutral source-context components; parenthetical values are numbers of free parameters. Estimating the multiplier tested the directional hypothesis of preferential emotional learning while also permitting attenuation.

Early-LPP modulation	Emotional-learning multiplier	Emotional-learning multiplier
	fixed at 1	estimated
None	Source-context baseline (11)	Preferential-learning model (12)
General	General-LPP model (12)	Preferential learning + General-LPP model (13)
Emotion-dependent	Emotion-dependent-LPP model (12)	Preferential learning + Emotion-dependent-LPP model (13)

Table 2*Parameters in the six-model specification.*

Parameter	Role	Fitted range or fixed value
β_{enc}^T	Temporal-context drift during study	(0, 1)
β_{enc}^S	Source-context drift during study	(0, 1)
β_{start}	Drift toward start-of-list context before recall	(0, 1)
β_{rec}^T	Temporal-context reinstatement after recall	(0, 1)
β_{rec}^S	Source-context reinstatement after recall	1 (fixed)
α	Shared pre-experimental context-to-item support	(0, 100]
δ	Item-specific pre-experimental context-to-item support	(0, 100]
γ	Feature-to-context learning rate	(0, 1)
ϕ_s	Primacy scale	(0, 100]
ϕ_d	Primacy decay	(0, 100]
τ_c	Recall-choice sensitivity	(0, 100]
ω	Source learning relative to temporal learning	(0, 10]
ϕ_{emot}	Emotional-learning multiplier for negative items	1 (fixed) or (0, 10]
κ	General Early-LPP slope on log learning strength	0 (fixed) or [0, 0.2145]
κ_{emot}	Negative-item Early-LPP slope	0 (fixed) or [0, 0.2145]

Table 3

Pooled fit of the six eCMR model variants. The first two columns identify which learning mechanisms were available for estimation; they do not indicate whether either mechanism improved fit. p is the number of free parameters; NLL is the best-found negative log-likelihood across three differential-evolution starts; ΔAIC is relative to the lowest-AIC model. Bold values identify the best-fitting model.

Emotion-based learning boost	LPP-based learning boost	p	NLL	AIC	ΔAIC
No	None	11	8312.401	16646.803	136.826
No	All items	12	8300.416	16624.832	114.855
No	Negative items only	12	8283.055	16590.109	80.133
Yes	None	12	8257.154	16538.309	28.332
Yes	All items	13	8251.272	16528.545	18.568
Yes	Negative items only	13	8241.988	16509.977	0.000

Table 4

Observed and model-predicted recall-rate and original-z Early-LPP contrasts. Parentheses give prediction minus observed; predictions are means across 200 complete simulated datasets. The Early-LPP contrasts were not separately fitted.

eCMR specification	Early LPP:			
	Negative items			
	(Remembered –			
	Forgotten) – Neutral items			
	Recall rate: Negative – Neutral	Early LPP: Remembered – Forgotten, Negative items	Early LPP: Remembered – Forgotten, Neutral items	Early LPP: Remembered – Forgotten, Neutral items
Observed data	0.124	0.455	0.035	0.420
Baseline	0.000 (–0.124)	0.029 (–0.425)	0.009 (–0.026)	0.020 (–0.400)
All-item LPP boost	0.006 (–0.117)	0.240 (–0.215)	0.207 (+0.172)	0.033 (–0.387)
Negative-item LPP boost	0.019 (–0.105)	0.560 (+0.105)	0.020 (–0.015)	0.539 (+0.120)
Emotion-based boost	0.126 (+0.002)	0.028 (–0.426)	0.010 (–0.024)	0.018 (–0.402)
Emotion-based + all-item LPP boosts	0.140 (+0.016)	0.330 (–0.125)	0.066 (+0.031)	0.264 (–0.156)
Emotion-based + negative-item LPP boosts	0.119 (–0.004)	0.447 (–0.007)	0.024 (–0.011)	0.424 (+0.004)

Table 5*Key for the six primary model columns in Table 6.*

Code	Emotional-learning		Manuscript label
	multiplier	Early-LPP mapping	
A	Fixed at 1	None	Source-context baseline
B	Fixed at 1	General	General-LPP model
C	Fixed at 1	Emotion-dependent	Emotion-dependent- LPP model
D	Estimated	None	Preferential-learning model
E	Estimated	General	Preferential learning + General-LPP model
F	Estimated	Emotion-dependent	Preferential learning + Emotion- dependent-LPP model

Table 6

Best-found pooled parameter estimates for the six primary models. Values are rounded for presentation; square brackets identify fixed values. Parameter roles and fitted ranges are reported in Table 2.

Parameter	A	B	C	D	E	F
β_{enc}^T	0.71851	0.74573	0.75492	0.82721	0.0049261	0.7364
β_{enc}^S	0.060535	0.034311	0.017177	0.056109	0.19086	0.059001
β_{start}	0.99726	0.99987	0.99998	1	0.030802	0.99996
β_{rec}^T	0.42503	0.094411	0.17007	0.096094	0.18614	0.10575
β_{rec}^S	[1]	[1]	[1]	[1]	[1]	[1]
α	0.30498	20.437	6.2235	10.59	94.457	5.8256
δ	23.84	65.123	31.432	64.685	22.825	58.887
γ	0.2887	0.54427	0.30069	0.32279	0.34882	0.28719
ϕ_s	1.3389	0.14933	0.16018	0.31836	3.943	0.33536
ϕ_d	82.54	2.0841	22.287	44.051	1.2901	76.191
τ_c	0.64638	5.0889	3.6255	2.5134	1.8339	1.9979
ω	1.8797	6.0171	5.8759	2.4608	3.2044	2.6692
ϕ_{emot}	[1]	[1]	[1]	1.6302	5.6737	1.5774
κ	[0]	0.032287	[0]	[0]	0.15144	[0]
κ_{emot}	[0]	[0]	0.099237	[0]	[0]	0.11542

Figure 1

Observed and model-predicted recall rates for negative and neutral items. Brackets show the negative-minus-neutral difference. Observed error bars are 95% participant-cluster bootstrap confidence intervals; predicted error bars are central 95% intervals across 200 complete simulated datasets.

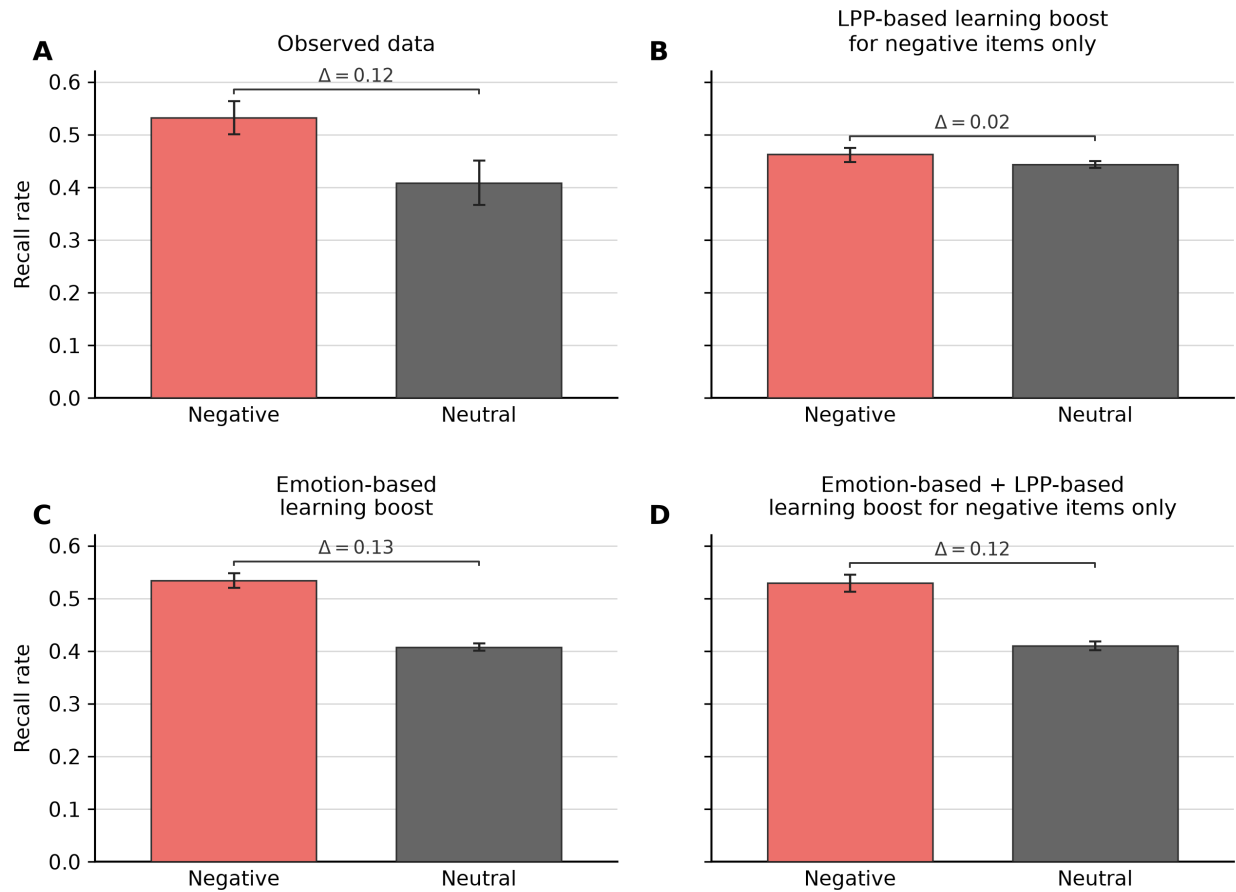


Figure 2

Observed and model-predicted original z-transformed Early LPP by item category and memory status. Brackets show remembered-minus-forgotten differences within each category. Observed error bars are 95% participant-bootstrap confidence intervals; predicted error bars are central 95% intervals across 200 complete simulated datasets.

